

EXAMPLES

STEP 1: CREATE PHYSICIAN TABLES

```
[mysql> USE student_db
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> CREATE TABLE physician (
  ->     employeeid INT PRIMARY KEY,
  ->     name VARCHAR(100),
  ->     position VARCHAR(100),
  ->     ssn VARCHAR(20)
[  -> );
Query OK, 0 rows affected (0.018 sec)
```

STEP 2: INSERT DATA INTO PHYSICIAN

```
mysql> INSERT INTO physician VALUES
  -> (1,'John Dorian','Staff Internist','111111111'),
  -> (2,'Elliot Reid','Attending Physician','222222222'),
  -> (3,'Christopher Turk','Surgical Attending Physician','333333333'),
  -> (4,'Percival Cox','Senior Attending Physician','444444444'),
  -> (5,'Bob Kelso','Head Chief of Medicine','555555555'),
  -> (6,'Todd Quinlan','Surgical Attending Physician','666666666'),
  -> (7,'John Wen','Surgical Attending Physician','777777777'),
  -> (8,'Keith Dudemeister','MD Resident','888888888'),
[  -> (9,'Molly Clock','Attending Psychiatrist','999999999');
Query OK, 9 rows affected (0.003 sec)
Records: 9 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM physician;
+-----+-----+-----+-----+
| employeeid | name           | position                | ssn       |
+-----+-----+-----+-----+
| 1          | John Dorian    | Staff Internist         | 111111111 |
| 2          | Elliot Reid    | Attending Physician     | 222222222 |
| 3          | Christopher Turk | Surgical Attending Physician | 333333333 |
| 4          | Percival Cox   | Senior Attending Physician | 444444444 |
| 5          | Bob Kelso      | Head Chief of Medicine   | 555555555 |
| 6          | Todd Quinlan   | Surgical Attending Physician | 666666666 |
| 7          | John Wen       | Surgical Attending Physician | 777777777 |
| 8          | Keith Dudemeister | MD Resident             | 888888888 |
| 9          | Molly Clock    | Attending Psychiatrist   | 999999999 |
+-----+-----+-----+-----+
9 rows in set (0.000 sec)
```

STEP 3: CREATE DEPARTMENT TABLE

```
mysql> CREATE TABLE department (
  ->     departmentid INT PRIMARY KEY,
  ->     name VARCHAR(100),
  ->     head INT
[  -> );
Query OK, 0 rows affected (0.008 sec)
```

STEP 4: INSERT DATA INTO DEPARTMENT

```
mysql> INSERT INTO department VALUES
      -> (1,'General Medicine',4),
      -> (2,'Surgery',7),
      -> (3,'Psychiatry',9);
Query OK, 3 rows affected (0.003 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
[mysql> SELECT * FROM department;
```

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9

```
3 rows in set (0.000 sec)
```

QUESTION 1:- INNER JOIN

```
mysql> SELECT d.name AS Department,
      -> p.name AS Head_Physician
      -> FROM department d
      -> JOIN physician p
      -> ON d.head = p.employeeid;
```

Department	Head_Physician
General Medicine	Percival Cox
Surgery	John Wen
Psychiatry	Molly Clock

```
3 rows in set (0.001 sec)
```

QUESTION 2:- WHERE CLAUSE

```
mysql> SELECT *
      -> FROM physician
      -> WHERE position = 'Surgical Attending Physician';
```

employeeid	name	position	ssn
3	Christopher Turk	Surgical Attending Physician	333333333
6	Todd Quinlan	Surgical Attending Physician	666666666
7	John Wen	Surgical Attending Physician	777777777

```
3 rows in set (0.001 sec)
```

QUESTION 3:- WHERE + LIKE

```
mysql> SELECT *
-> FROM physician
-> WHERE name LIKE 'John%';
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
7	John Wen	Surgical Attending Physician	777777777

2 rows in set (0.002 sec)

QUESTION 4:- COUNT()

```
mysql> SELECT COUNT(*) AS Total_Physicians
-> FROM physician;
```

Total_Physicians
9

1 row in set (0.001 sec)

QUESTION 5:- COUNT DISTINCT

```
mysql> SELECT COUNT(DISTINCT position)
-> FROM physician;
```

COUNT(DISTINCT position)
7

1 row in set (0.001 sec)

QUESTION 6:- GROUP BY

```
mysql> SELECT position,
-> COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position;
```

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

7 rows in set (0.001 sec)

QUESTION 7:- GROUP BY + HAVING

```
mysql> SELECT position,  
->      COUNT(*) AS Total_Employees  
-> FROM physician  
-> GROUP BY position  
-> HAVING COUNT(*) > 1;
```

position	Total_Employees
Surgical Attending Physician	3

1 row in set (0.001 sec)